

GANGCA, China

WMO: 527540

Lat: 37.3305N

Lon: 100.1380E

Elev: 3302

StdP: 67.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-19.4	-33.2	0.3	-14.2	-31.0	0.3	-14.0	12.2	-4.5	10.8	-4.9	2.1	0	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	20.2	11.6	18.7	10.8	17.2	9.9	13.0	18.4	12.1	17.0	11.1	15.7	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	12.4	15.8	10.1	11.6	14.9	9.1	10.9	14.1	48.7	18.6	45.5	17.1	42.6	15.9	16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.7	8.9	7.4	DB	-24.8	22.9	2.0	1.7	-26.3	24.1	-27.4	25.1	-28.5	26.0	-30.0	27.3	(r)
			WB	-25.6	14.2	1.8	1.0	-26.9	15.0	-28.0	15.5	-29.0	16.1	-30.3	16.8	(s)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-11.5	-8.4	-3.6	2.2	6.1	9.3	12.0	11.6	7.9	2.0	-4.4	-9.4
	DBStd	8.58	3.07	3.55	3.59	3.39	2.66	2.26	2.44	2.50	2.56	2.81	3.11	2.61
	HDD10.0	3370	667	515	420	233	124	38	7	11	75	249	431	601
	HDD18.3	6251	925	748	678	483	379	271	197	208	314	507	681	859
	CDD10.0	162	0	0	0	1	3	17	69	62	11	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.2	2.6	3.0	3.6	3.8	3.7	3.3	3.1	3.0	2.9	2.9	3.1	2.9
Precipitation	PrecAvg	379	1	2	5	13	40	69	87	84	55	15	3	1
	PrecMax	516	5	5	20	36	90	115	157	146	127	42	13	4
	PrecMin	260	0	0	0	1	3	28	31	45	17	0	0	0
	PrecStd	63	1	1	4	10	21	22	25	25	22	11	3	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	4.9	10.9	15.8	17.6	19.3	23.4	22.0	18.5	13.4	7.2	2.3
		MCWB	-5.6	-3.2	0.3	3.2	5.5	9.7	12.8	12.5	11.3	4.4	-0.9	-4.1
	2%	DB	-1.5	2.6	7.7	13.1	15.5	17.5	20.8	20.1	16.2	11.2	5.5	0.0
		MCWB	-6.9	-4.2	-1.2	2.5	5.0	8.9	12.0	11.7	9.6	3.2	-1.9	-5.8
	5%	DB	-3.2	0.7	5.6	11.1	13.9	16.0	19.0	18.6	14.5	9.6	4.0	-1.5
		MCWB	-8.0	-5.4	-2.4	1.6	4.4	8.1	11.1	11.3	8.2	2.1	-2.8	-6.7
	10%	DB	-4.9	-1.3	3.6	9.2	12.3	14.7	17.3	17.0	13.1	8.1	2.4	-3.2
		MCWB	-9.1	-6.5	-3.2	0.7	3.9	7.5	10.4	10.5	7.1	1.3	-3.7	-7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.4	-2.8	0.9	4.9	7.7	12.2	14.4	14.3	12.5	6.6	0.0	-3.9
		MCDB	0.0	4.1	9.0	12.6	14.0	17.2	20.6	19.7	16.6	11.5	5.6	1.6
	2%	WB	-6.8	-4.1	-0.8	3.4	6.5	10.4	13.1	13.1	10.8	4.5	-1.5	-5.4
		MCDB	-1.8	2.3	6.3	10.9	12.8	15.2	19.0	18.2	14.8	9.1	4.5	-0.3
	5%	WB	-7.9	-5.2	-1.8	2.4	5.6	9.3	12.1	12.2	9.4	3.2	-2.5	-6.5
		MCDB	-3.4	0.3	4.6	9.6	11.6	14.0	17.3	17.0	13.4	8.0	3.3	-1.9
	10%	WB	-9.0	-6.4	-2.9	1.4	4.7	8.4	11.2	11.2	8.1	2.0	-3.4	-7.6
		MCDB	-5.1	-1.6	3.1	8.0	10.6	13.3	16.0	15.7	12.0	6.9	1.9	-3.3
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.1	12.8	11.4	10.3	9.3	9.1	9.3	9.5	11.3	12.6	12.7
		MCDBR	13.9	14.6	14.8	13.9	12.9	11.5	11.6	11.0	10.8	12.7	13.3	13.6
		MCWBR	10.2	9.9	8.7	6.8	6.1	5.6	5.3	5.2	5.6	6.6	8.1	9.6
	5% WB	MCDBR	13.5	14.0	13.1	12.1	10.5	9.2	9.3	8.9	9.1	10.6	12.5	13.0
		MCWBR	10.0	9.6	7.8	6.3	5.6	5.0	4.7	4.7	5.0	6.1	7.9	9.4
Clear-Sky Solar Irradiance	taub	0.212	0.249	0.304	0.357	0.373	0.374	0.363	0.358	0.325	0.275	0.228	0.203	
	taud	2.469	2.383	2.245	2.121	2.163	2.248	2.357	2.356	2.417	2.493	2.500	2.500	
	Ebn at Noon	1014	998	960	920	907	903	911	909	927	956	982	1006	
	Edh at Noon	79	99	127	153	150	138	122	120	106	88	76	71	
All-Sky Solar Radiation	RadAvg	3.11	3.90	5.00	5.92	6.19	5.99	5.92	5.46	4.62	4.16	3.36	2.90	
	RadStd	0.18	0.18	0.26	0.25	0.32	0.33	0.45	0.36	0.30	0.24	0.20	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.44	N/A	-1.19	N/A	N/A	N/A	-113	-160	+18	N/A
(47)	Regional (2 neighbors)	+0.33	N/A	-0.80	N/A	N/A	N/A	N/A	N/A	+52	N/A

Nomenclature: See separate page